Giants vs. Dwarfs: Most Abundant Star in the Universe

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Abstract

Stars are the main component of the cosmos and studying their masses will help us understand their formation and evolution. We use data from the Gaia telescope to select potential members in a stellar cluster of Orion. We derive their masses using evolutionary models and create an initial mass function (IMF) to find the mass of the most common star in the stellar group. **The shape of our IMF matches that of other clusters, meaning that this trend could be applied universally**, allowing us to infer the most abundant star in the universe.

Introduction

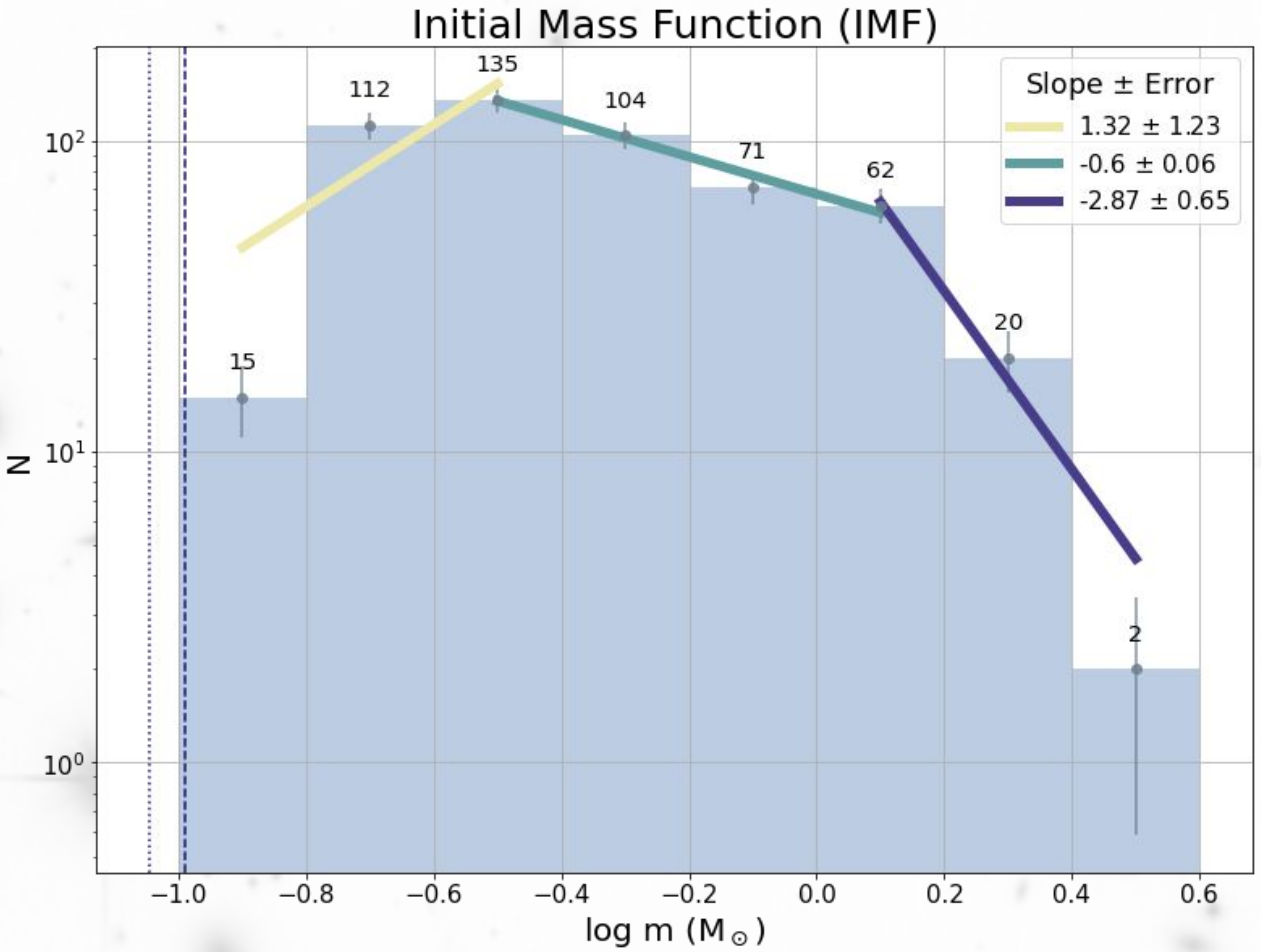
Our universe is composed of "bright little lights", which are in fact giant balls of gas called stars. The stars we observe at night are not identical, they have different colors and brightnesses as a result of the different amounts of gas (mass) they accumulated during their formation. Mass is the most important parameter of a star because it allows us to understand a star's life cycle. **Focusing on the Orion star-forming complex, we study the mass distribution of stars in a cluster to determine the most common type of star** and interpret our findings in a broader context.

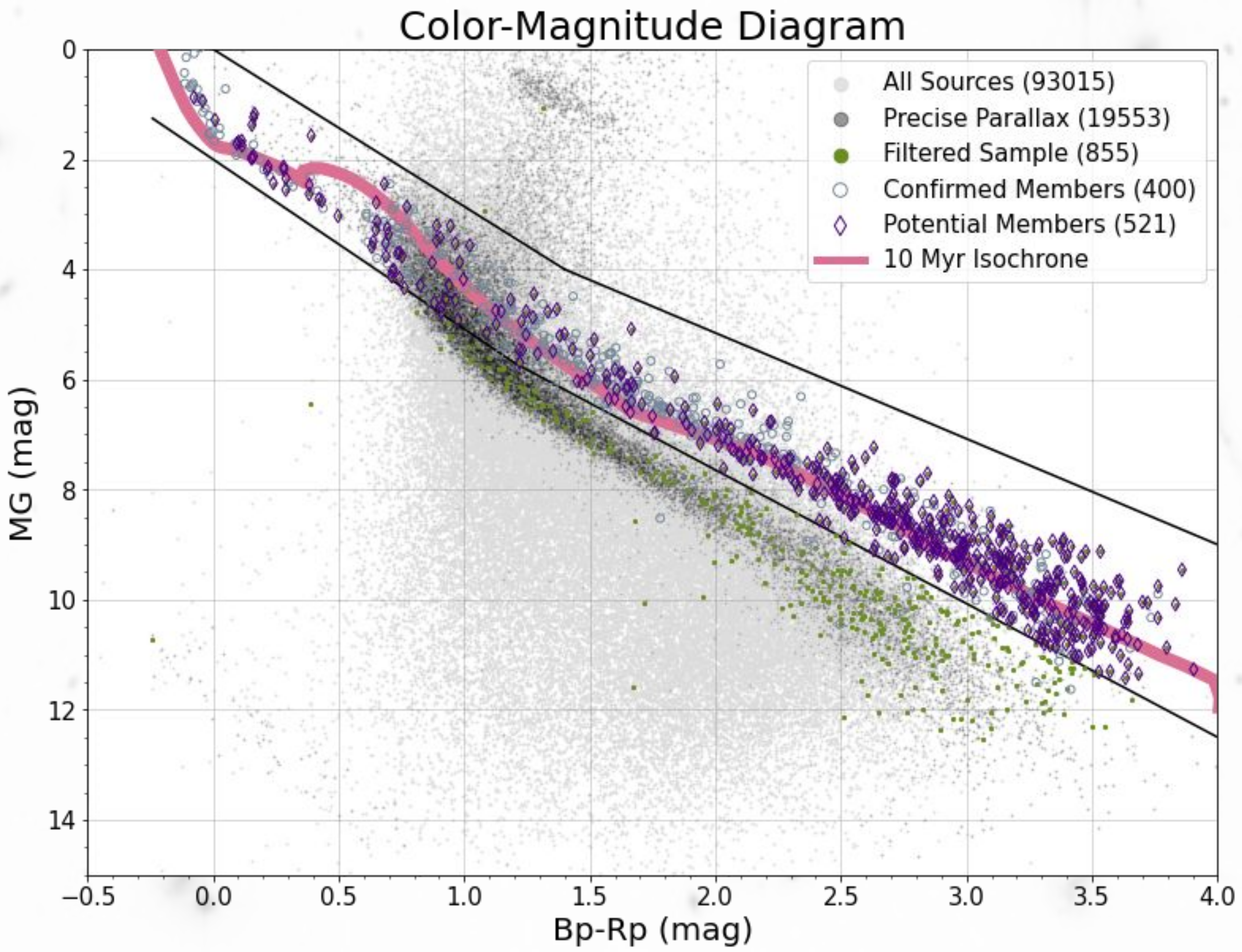
Methods

Using the Gaia space telescope, we first retrieve data of stars towards the Orion D stellar association. **We use these data to select potential members of the group applying by several criteria** (e.g. K18, S19):

- Consistent distance with the group (200-500 pc; < 10% error)
- Moving with the group (-4<=Proper Motion<=4)
- Exclude binaries (RUWE<1.4)
- Keep young stars (using a Color-Magnitude M_G vs Bp-Rp diagram)

We then determined masses for the selected stars by using evolutionary models (M17), assuming the group's age (10 Myr; K18). These masses were used to create a mass distribution of the star cluster, known as an Initial Mass Function (IMF).





Results

Our final sample has 521 potential members ranging from 0.13 to 2.8 $M_\odot$ and a median mass of 0.40 $M_\odot$. **Our IMF peaks at 0.32 $M_\odot$.**

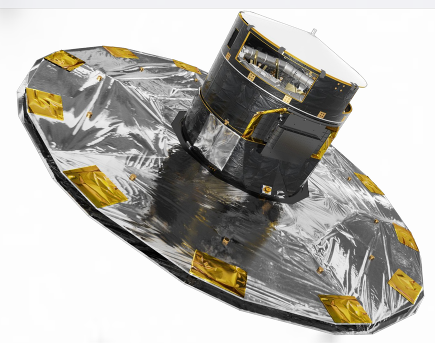
IMF; $\xi(\log m) = (\Delta N/\Delta \log m) \propto m^{\Gamma}$	
Mass Range ($M_\odot$)	**Slope (Γ)**
0.1 - 0.3	1.32±1.23
0.3 - 1.3	-0.6±0.06
>0.3	-2.87±0.65

Discussion

We found that the peak of the IMF is at ≈0.3 $M_\odot$, meaning that the most common type of star is 30% the mass of our Sun. The resulting IMF is similar to those found in other stellar associations (C23, S19), suggesting that **the most abundant star in our universe is most likely 0.3 $M_\odot$.**

References

- C23: Cordoni et al. 2023, A&A, accepted
- K18: Kounkel et al. 2018, AJ, 156, 84
- M17: Marigo et al. 2017, A&A, 391, 195
- S19: Suárez et al. 2019, MNRAS, 486, 1718



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